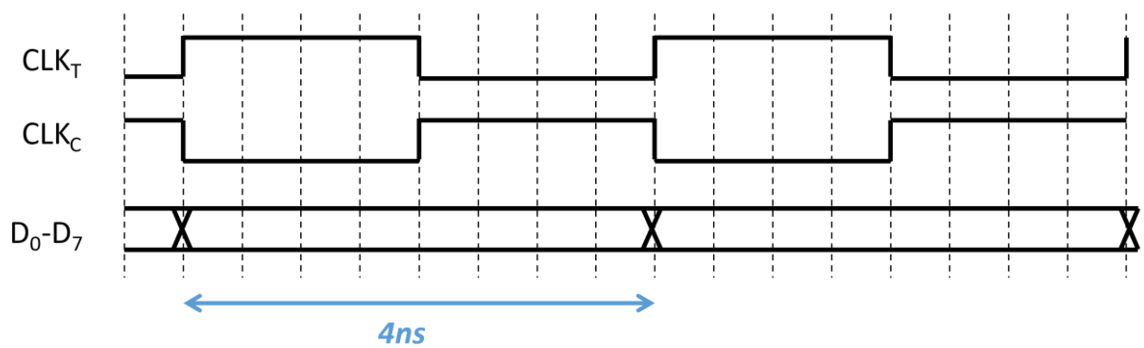
ELEN E6316
Spring 2026
Tod Dickson **8b DAC Design Project (v1.0)** Due: Friday May 1, 2026

DESIGN SPECIFICATIONS

High-speed DACs are often used as output drivers in wireline communication transmitters. The use of a DAC makes the transmitter flexible for a number of wireline applications, as it enables support for a wide variety of modulation and equalization schemes.

You are to design a high-speed 250MS/s 8b DAC in a 0.18μm CMOS technology, operating from a 1.8V supply voltage. The DAC should provide differential outputs with a maximum output swing of at least 1.5V peak-to-peak differential (750mV per side), with an SFDR of better than 48dB across the Nyquist band. The differential DAC, which is intended to drive a differential transmission line channel with 100Ω characteristic impedance, should have a 100Ω differential output resistance and a 25Ω common-mode output resistance. Moreover, parasitic capacitances associated with the output bond pad and electrostatic discharge (ESD) protection devices introduce an additional 550fF of capacitive loading to ground at each of the two outputs of the differential DAC. The DAC should have DNL better than +/- 1 LSBs and INL better than +/- 2 LSBs. To combat random variations , it is required that the analog weighting of each DAC bit have +/- 10% tuning range. The analog weighting of each DAC bit can be tuned via independent 6b DACs. A single 100μA reference current is available for use. Any additional bias currents must be derived from this 100μA reference.

The single-ended input 8b digital data along with a differential 250-MHz clock is driven by CMOS inverters with PMOS W/L = 720nm/180nm, and NMOS W/L = 360nm/180nm. The data is aligned to the rising edge of the true phase of the 250-MHz clock (CLKT), as showing in the timing diagram below. Assume that the data is unsigned.



DELIVERABLES

Work in groups of two. Each group is responsible for delivering a top-level DAC schematic, as well as all required sub-macros or building blocks. Schematic entry is to be done in the Cadence design environment, with simulations performed in the Cadence Analog Design Environment. One report per group should be handed in, as well as a path to Cadence library and the names of top-level schematics and relevant test benches. It is expected that schematics should be organized in a logical hierarchy, and that higher-level schematics make use of symbols to instantiate lower-level building blocks or sub-macros.

TECHNOLOGY & MODEL FILES

Transistor models for the 0.18μm CMOS technology can be found at

http://www.ee.columbia.edu/~kinget/TOOLS/technology.html

You will only be using the transistor models found in the TSMC018_teaching.scs file. You do not need access to any Process Development Kit (PDK) and do not need to sign any Non-Disclosure Agreement (NDA). This design project does not involve any physical design (layout).

Simulations should be performed over process corners. In the model file, the tt section corresponds to a typical n/typical p process corner. The ss section corresponds to a slow corner while the ff section corresponds to a fast corner. However, these corners do not include device mismatch. Devices that are sensitive to mismatch should be sized accordingly. As the model file does not include device mismatch parameters, you can use the following for performing hand-calculations of component sizes: $A_{VT} = 5mV*\mu m$, $A_{\beta} = 0.01\mu m$.

You are only using the TSMC018_TEACHING.SCS model file, not the TSMC technology library. You do not need to set up the TSMC018 technology as described on the webpage, and an NDA is not required. You can instantiate MOSFETs from the "analogLib" library using the "nmos4" or "pmos4" devices, and typing in the appropriate model name (nch or pch) into the properties field for each device. Ideal resistors and capacitors can be used. However, for each capacitor you should include a 10% bottom plate capacitor to ground.

WORKING IN PAIRS

You are supposed to work in pairs. If you do not have a partner, email the instructor as soon as possible. You need only hand in one report per pair. Both partners are expected to contribute about equally to the effort. If this is not the case for you and your partner, *you are asked to state, on the first page of your report, the approximate percentage of each partner's contribution*.

REPORT

Your typed report should be **NO MORE THAN 7 PAGES** and should include (but is not limited to) the following:

1. *Architecture:* Give a high-level overview of the DAC architecture. Why was the architecture chosen? Clearly describe how the DAC analog weight tuning is accomplished.
2. *Building blocks:* Describe the building blocks (circuit components) in your DAC architecture. Make connections between DAC specifications and building block design requirements. Where appropriate, include relevant biasing information.
3. *DAC design and simulation*. Include a top-level schematic and describe its operation. Describe the function of each block on a given clock cycle. You should simulate the relevant DAC static and dynamic performance.
   a. DAC static performance: Simulate the INL and DNL. Demonstrate the tuning range of the DAC analog weights.
   b. DAC dynamic performance: Dynamic performance (SFDR, SNDR) can be simulated by running transient simulations of the DAC. Its inputs should be configured in order to produce an output sinusoid at a desired frequency. You can then take an FFT of the output waveform to examine the spectrum. The exact frequency and length of the transient simulation must be chosen carefully. Large number of DAC output samples (hence, long transients) are required such that the FFT will have sufficiently fine frequency resolution. Additionally, the relationship between the output frequency and the sampling rate must be chosen to avoid deterministic quantization noise in the output waveform. You should plot SNDR and SFDR as a function of input frequency. The Cadence calculator has a built-in discrete Fourier transform (DFT) function, which also includes windowing capabilities. The instructor will post some notes and a video with more details on setting up these simulations.

   Describe your test benches used to simulate these performance metrics. Summarize your DAC performance in a table. Include power dissipation, INL/DNL, and SNDR/SFDR at low and high frequencies. A brief discussion on performance limitations (what block or blocks do you feel is/are limiting your SNDR/SFDR the most) is also expected. Break down the power dissipation by circuit block.

Note that Cadence schematics are of poor quality and are unacceptable in the report. Cadence plots of simulation results are acceptable provided you make an effort to reasonably annotate your results.

A report template following IEEE journal formatting is posted on Courseworks.

GRADING

The project will be graded based on the quality of the design as well as the quality of the written report. As a design engineer, it is insufficient just to put some transistors together in a simulation and demonstrate that it works. You are expected to know *why* the design works, and you should be able to support this with relevant analysis. Hence, the report should include significant analysis of your circuit in addition to the simulation results. ***Design creativity is encouraged!*** The report should be completely self-explanatory; there should not be a need to submit additional files.

DUE DATE: Friday, May 1, 2026. Submit your design report in electronic format through Canvas. Do not submit hard copies. Late reports will **not** be accepted.